

Solution

GRAVITY-04

Class 09 - Science

Section A

1.

(b) Both A and R are true but R is not the correct explanation of A.
Explanation: Since, $g = G \frac{M}{R^2}$, the acceleration due to gravity depends both on average density and the average radius of the earth.
2.

(b) Both A and R are true but R is not the correct explanation of A.
Explanation: Net force = real weight - upthrust (weight of the liquid displaced).
The object rises above the liquid surface to an extent that the weight of the liquid displaced by the part of the object. Thus, an object will float when upward thrust is greater than its actual weight. (The density of the object must be less than the density of the liquid).
3.

(c) A is true but R is false.
Explanation: The gravitational force of attraction is the reason behind the formation of tides. It is a weak force.
4.

(c) A is true but R is false.
Explanation: The mass of the body is the quantity of matter contained in it. It is a scalar quantity that has only magnitude but no direction.
5.

(a) Both A and R are true and R is the correct explanation of A.
Explanation: A school bag has wide straps so that the weight of the books may spread over a large area of a child's shoulder producing less pressure.
6.

(a) Both A and R are true and R is the correct explanation of A.
Explanation: The change in direction involves a change in velocity or acceleration. The force that causes this acceleration and keeps the body moving along the circular path is acting towards the centre. This force is called the centripetal force.
The motion of the moon around the earth is due to the centripetal force.
7.

(c) A is true but R is false.
Explanation: Weight of an object is the force with which a body is attracted towards the earth. Its direction is vertically downwards.
8.

(a) Both A and R are true and R is the correct explanation of A.
Explanation: The falling of a body from a height towards the earth under its gravitational force is called free fall. So, whenever objects fall towards the earth under this force alone, we say that the objects are in free fall.
9.

(c) A is true but R is false.
Explanation: A is true but R is false.
10.

(b) Both A and R are true but R is not the correct explanation of A.
Explanation: The value of g changes with height, depth and is zero at the centre of the earth.
11.

(c) A is true but R is false.
Explanation: A is true but R is false.
12.

(a) Both A and R are true and R is the correct explanation of A.
Explanation: Since the earth's mass is much more than that of the moon, the gravitational force exerted by the earth on the spaceship is much greater than that exerted by the moon. The work done is much greater.

13. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: According to the universal law of gravitation, every object in the universe attracts every other object with a force that is proportional to the product of their masses and inversely proportional to the square of the distance between them. The direction of force is along the line joining the centres of the two bodies.
14. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
15. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: $F = G \frac{m_1 m_2}{r^2}$
16. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: The value of G is independent of direction and is a constant through out the universe.
17. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: The weight of an object on the moon will be about one-sixth of what is on the earth. Thus, a spring balance that shows the weight of a body to be 6 N on earth will show a weight of only 1 N when taken to the moon.
18. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
19. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: According to Kepler's law, $T^2 \propto r^3$ or $T \propto r^{\frac{3}{2}}$
 Where r becomes four times. T is $t^{\frac{3}{2}} (=2^3) = 8$ times
20. **(d)** A is false but R is true.
Explanation: The acceleration experienced by an object during free fall is independent of its mass. This means that all objects hollow or solid, big or small, should fall at the same rate.
21. **(d)** A is false but R is true.
Explanation: A is false but R is true.
22. **(c)** A is true but R is false.
Explanation: The force of attraction of the earth on an object is known as the weight of the object. It is denoted by W. The SI unit of weight is Newton (N).
23. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: The weight of an object on the moon is less than that on the earth because the mass and radius of the moon are less than that of earth.
24. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: At the centre of the earth value of g is zero, weight of the body is also zero but its mass remains the same. The centre of mass of the body thus remains unchanged.
25. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: When a piece of cork is held below the surface of the water by applying the force of our thumb and then released, the cork immediately rises to the surface because the upward force is exerted by water on the cork which pushes it to the surface.
26. **(c)** A is true but R is false.
Explanation: A is true but R is false.

Section B

27. State True or False:

- (i) **(a)** True
Explanation: True

- (ii) **(a)** True
Explanation: True
- (iii) **(a)** True
Explanation: True
- (iv) **(a)** True
Explanation: True
- (v) **(a)** True
Explanation: True
- (vi) **(a)** True
Explanation: True
- (vii) **(a)** True
Explanation: True
- (viii) **(a)** True
Explanation: True
- (ix) **(a)** True
Explanation: True
- (x) **(a)** True
Explanation: True
- (xi) **(a)** True
Explanation: True
- (xii) **(a)** True
Explanation: True
- (xiii) **(b)** False
Explanation: False. An object with a density less than the liquid floats on the liquid.
- (xiv) **(b)** False
Explanation: False. The value of 'g' is maximum on the surface of the earth.
- (xv) **(b)** False
Explanation: False. Kepler's Third Law states that the cube of the mean distance 'r' of a planet from the Sun is directly proportional to the square of its orbital period 'T'.
- (xvi) **(a)** True
Explanation: True
- (xvii) **(a)** True
Explanation: True

Section C

28. Fill in the blanks:

- (i) 1. One-sixth
- (ii) 1. Buoyant force
- (iii) 1. Gravity
- (iv) 1. Earth
- (v) 1. Acceleration
- (vi) 1. Float
- (vii) 1. Floats
- (viii) 1. Centripetal
- (ix) 1. Mass
- (x) 1. Thrust
- (xi) 1. Newton
- (xii) 1. Gravitational
- (xiii) 1. Weight

- (xiv) 1. Force
(xv) 1. Pressure
(xvi) 1. Mass

Section D

29.
(c) 40 g wt
Explanation: Loss in weight = Difference in reading, i.e. $84 - 44 = 40$ units. [Scale is not graduated with respective units]
30.
(b) All of these
Explanation: The overflow can must always be filled upto its spout before using it to measure the volume displaced by the immersed solid.
31.
(c) 2 g/cm^3
Explanation: $Density = \frac{mass}{volume} = \frac{8}{9-5} = \frac{8}{4} = 2 \text{ g/cm}^3$
32.
(d) Same in all
Explanation: Loss in weight depends on the liquid density and volume displaced (= volume immersed) and not on the shape of the container.
33.
(c) 20 cm^3
Explanation: $60 - 40 = 20 \text{ cm}^3$
34.
(a) 5 gwt
Explanation: The reading on the spring balance when suspended freely in the air.
35.
(d) C
Explanation: Water has to be filled to the level of overflow.
36.
(b) $U_w > U_s$
Explanation: The density of salty water is more as compared to tap water hence, the upthrust by salty water is more.
37.
(b)
1.8

Explanation: Volume of block = change in level by volume = $(4.2 - 2.4) = 1.8 \text{ cm}^3$

38.
(b) 2 g
Explanation: $L.C. = \frac{20 - 0}{10} = \frac{20}{10} = 2 \text{ g}$

Section E

39.
(c) Statement B
Explanation: The value of g decreases as we move at a higher altitude, so 'g' is less at Ooty than Delhi. Weight will be less in Ooty than Delhi. Mass is measured by pan of beam balance and weight is measured by spring balance.
40.
(d) statement A
Explanation: The radius of the earth at the poles is 6357 km, the radius at the equator is 6378 km.
The force by which earth attracts an object towards it is called the weight of the object, which is the product of mass (m) of the object and acceleration due to gravity (g). Mass is the measurement of inertia. So, at a given place the weight of an object is not

equal to the mass of the object.

So, Statement A is true and B is false.

41. **(a)** neither statement A nor B.

Explanation: Earth and moon attract each other with equal gravitational force. The mass of a body never becomes 0. Hence both statements are incorrect.

42. **(a)** Statement A

Explanation: Mass of a body always remains the same. Mass is independent of other factors. Hence mass is a fundamental property. On the other hand weight of the body is the force by which it is attracted by the earth that depends upon the value of 'g'. So, weight is not a fundamental factor.

43. **(a)** Both A and B

Explanation: When an object is thrown up its speed decreases with height as gravitational force acting on it is negative. After attaining maximum height objects start falling downward with increasing speed as the gravitational force acting is positive.

44. **(a)** Statement B

Explanation: The force of attraction is always attractive. The law of the inverse square means that the force by which two objects attract each other is inversely proportional to the square of the distance between them.